Software Test Plan (STP)

Enterprise Conference Room Booking System

Project: Enterprise Conference Room Booking System (CRB)
Version: 0.1
Authors: **SINCHANA K PES1UG24AM405**
DIRAN SOMANNA PES1UG24AM364
DISHANTH GOWDA H D PES1UG24AM365
PAVAN D C PES1UG24AM383
Date: 5-10-2026

Revision History

Version	Date	Author	Change Summary
0.1	29-09-2026	**SINCHANA K** **DIRAN SOMANNA** **DISHANTH GOWDA** **PAVAN D C**	Initial Test Plan draft derived from the CRB SRS and SAD

1. Introduction

Purpose: This document defines the test plan for the Enterprise Conference Room Booking System (CRB) v0.1. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing against the requirements in CRB_SRS.docx and the design in CRB_SAD.docx.

Scope: Testing covers employee-facing room search and booking, recurring bookings, booking modification/cancellation, admin room and equipment management, automated notifications, and utilization reporting. Corporate SSO's internal implementation and the employee directory's internal implementation are excluded - both are exercised only through their public API/mock in this system's test environment.

References: CRB_SRS.docx v0.1, CRB_SAD.docx v0.1, CRB_Architecture_Decisions.md, CRB_Requirements_Catalog.docx.

Definitions: CRB (Conference Room Booking), RBAC (Role-Based Access Control), SSO (Single Sign-On), TTL (Time To Live), RTM (Requirements Traceability Matrix), STRIDE (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege).

2. Test Items

- Authentication & Access Control module (Auth Service)
- Room Search module (Booking Service)
- Booking Creation module, including conflict detection and holds (Booking Service, Redis)
- Booking Modification module - reschedule, cancel, hold auto-release (Booking Service, Redis)
- Recurring Bookings module (Booking Service)
- Admin Room & Equipment Management module (Room Admin Service)
- Notifications module (Notification Service)
- Reporting module (Reporting Service)
- API Gateway / RBAC enforcement layer

3. Features to be Tested

Features mapped to SRS requirement IDs:

- CRB-F-001 to CRB-F-003: SSO authentication, RBAC, account lockout
- CRB-F-010 to CRB-F-012: Room search, equipment filter, availability exclusion
- CRB-F-020 to CRB-F-023: Conflict-free booking creation, temporary hold, attendees, confirmation
- CRB-F-030 to CRB-F-032: Reschedule, cancel, hold auto-release
- CRB-F-040 to CRB-F-041: Recurring bookings and single-occurrence edits
- CRB-F-050 to CRB-F-051: Room/equipment CRUD, maintenance block
- CRB-F-060 to CRB-F-062: Confirmation, reminder, cancellation notifications
- CRB-F-070: Room utilization report
- CRB-NF-001 to CRB-NF-006: Performance, availability, scalability, retention, accessibility, maintainability
- CRB-SR-001 to CRB-SR-006: TLS, RBAC enforcement, audit logging, session timeout, input validation, PII encryption

4. Features Not to be Tested

- Internal implementation of the corporate SSO/identity provider (tested only via its API contract, using a Keycloak sandbox as a stand-in)
- Internal implementation of the corporate employee directory (assumed correct; consumed read-only)
- Corporate SMTP gateway's own delivery infrastructure (tested via a local SMTP test double, e.g. MailHog, not the production mail server)
- Add-on features not selected for the current scope (e.g. calendar sync, QR check-in, IoT occupancy) - out of scope until an ADR and SRS/SAD update bring them in

5. Test Approach / Strategy

Levels

- Unit tests (module-level, one per component: Auth, Booking, Room Admin, Notification, Reporting)
- Integration tests (Booking Service to Redis lock/hold; Booking/Admin Service to PostgreSQL; Auth Service to SSO sandbox; Notification Service to SMTP test double)
- System tests (end-to-end CRB flows across the API Gateway)
- Acceptance tests (UAT with course coordinator / product owner sign-off)

Types

- Functional testing (all CRB-F requirements)
- Regression testing (re-run on every merge to main, given the trunk-based workflow)
- Performance testing (CRB-NF-001 response time, CRB-NF-003 concurrent-user load)
- Usability / accessibility testing (CRB-NF-005, WCAG 2.1 AA)
- Security testing (CRB-SR-001 to CRB-SR-006, detailed in 5.1)

Entry Criteria: CRB_SRS.docx and CRB_SAD.docx approved, a stable build deployed to the test environment, test data (rooms, users, roles) seeded, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open (in particular, zero known double-booking defects against CRB-F-020), all High-priority acceptance criteria satisfied.

5.1 Security Validation

- Verify TLS 1.2+ is enforced on every endpoint and plaintext connections are rejected (CRB-SR-001)
- Verify RBAC is enforced server-side by calling Admin-only endpoints directly with an Employee-role token (CRB-SR-002)
- Verify every admin action (room create/edit/deactivate, booking override/cancel) produces a complete audit log entry (CRB-SR-003)
- Verify session tokens expire after 30 minutes of inactivity (CRB-SR-004)
- Fuzz all booking and admin form fields with injection payloads (SQLi, XSS) and confirm rejection/sanitization (CRB-SR-005)
- Inspect the database directly to confirm PII columns (attendee names/emails) are encrypted at rest, not plaintext (CRB-SR-006)
- Penetration-test the authentication flow, including the local fallback path used during simulated SSO outages

6. Test Environment

Software: CRB backend (Spring Boot) v0.1, React web client, PostgreSQL test instance, Redis test instance, Keycloak SSO sandbox, MailHog SMTP test double.

Tools: Selenium or Playwright (UI automation), Postman (API testing), JMeter or k6 (performance/load testing), GitHub Issues (defect tracking, per the project's existing sprint board).

Test Data: seeded rooms across at least 2 simulated office locations, dummy employee/admin accounts across all three roles, pre-populated bookings to exercise conflict detection.

7. Test Schedule

Milestones (following SRS/SAD approval on 21-Sep-2026):

- Test case design: 22-Sep-2026 to 26-Sep-2026
- Test environment setup: 24-Sep-2026 to 27-Sep-2026
- Test execution start: 12-Ocy-2026
- Test execution end: 28-Oct-2026
- UAT: 29-Oct-2026 to 31-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts (Selenium/Playwright, Postman collections, JMeter/k6 scripts)
- Test Data (seed scripts)
- Test Execution Logs
- Defect Reports (GitHub Issues)
- Test Summary Report

9. Roles and Responsibilities

Role	Track	Responsibility
QA / Integration Lead	Test	Owns this Test Plan, the RTM, and the testability review gate on every FR before it moves to Done

Role	Track	Responsibility
Test Engineer (x2)	Test	Design and execute test cases across all levels, log defects, verify fixes
Auth & Access Lead	Dev	Support defect fixes and triage for Authentication, RBAC, and Security requirements
Booking Engine Lead	Dev	Support defect fixes and triage for Room Search, Booking Creation/Modification, and Recurring Bookings - highest-priority triage target given CRB-F-020's complexity
Room/Resource Admin Lead	Dev	Support defect fixes and triage for Admin Room Management
Product Owner	-	Approve test results and sign off on release readiness (Course Coordinator)

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery from the Dev track	Request early smoke builds; keep test case design (does not require a running build) ahead of execution
Test environment downtime (Postgres/Redis/Keycloak sandbox)	Maintain a backup environment on a cloud VM or container snapshot that can be restored quickly
Concurrency defects in booking conflict detection (CRB-F-020) surface late	Prioritize CRB-F-020/021/032 test cases first in the schedule, not last, given they are the system's highest-complexity requirements
Dependency on the SSO sandbox / SMTP test double behaving differently from the real corporate systems	Document known differences; re-verify SSO and notification flows against real systems during UAT
Only 4 team members across Dev and Test tracks limits parallel execution capacity	Sequence test design by module priority (Auth and Booking first) so partial coverage is still meaningful if time runs short

11. Assumptions & Dependencies

- CRB_SRS.docx and CRB_SAD.docx are approved before test case design begins
- A Keycloak SSO sandbox and MailHog SMTP test double are available and stable throughout test execution
- Test data (seed rooms, users, bookings) will be scripted and available before execution starts
- The four-team-member availability follows the same parallel Dev/Test structure established for the SRS sprint

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is too unstable to execute more than 30% of planned test cases
- A critical, unmitigated double-booking defect (CRB-F-020) is found - halt further booking-flow testing until fixed

Resume testing if:

- Blocking defects are resolved and verified
- The environment is confirmed stable

13. Test Case Management & Traceability

The RTM in CRB_SRS.docx Section 8 maps every requirement to its test case IDs. Sample mappings:

Req ID	Requirement (short)	Test Case(s)
CRB-F-001	SSO authentication	TC-Auth-01
CRB-F-020	Prevent double-booking	TC-Book-01
CRB-F-021	2-minute temporary hold	TC-Book-02
CRB-F-040	Recurring bookings	TC-Rec-01, TC-Rec-02
CRB-NF-001	Search response time	TC-Perf-01
CRB-SR-002	Server-side RBAC enforcement	TC-Sec-02

14. Test Metrics & Reporting

Metrics collected: % test cases executed, % passed/failed, defect density, defect aging, requirement coverage (against the RTM).

Reports: daily execution status during the execution window (29-Sep to 10-Oct), and a Final Test Summary Report before UAT begins.